

Homework Sheet — Stage 6: Data Preprocessing

All paths below are inside `homework/homework06/`.

Assignment

You will extend today's in-class work by writing modular cleaning functions, applying them to the provided raw dataset, and documenting your assumptions.

Instructions

1. Create functions in `src/cleaning.py`:
 - `fill_missing_median()`
 - `drop_missing()`
 - `normalize_data()`
2. Use the provided raw dataset in `data/raw/`.
3. In a Jupyter notebook:
 - Load dataset
 - Apply cleaning functions
 - Save cleaned dataset to `data/processed/`
 - Compare original vs cleaned data
 - Document all assumptions clearly
4. Update README with a section on cleaning strategy.

Explicit Chain

In the lecture, we learned how to **fill, drop, and scale** data systematically. Now, you will **adapt these methods into reusable functions and apply them to your dataset**.

Submission

- `src/cleaning.py`
- Preprocessing notebook
- Updated README
- Saved processed dataset

Grading Rubric

- **Correctness (40%)**: Functions work as expected
- **Documentation (20%)**: Clear assumptions, docstrings, README updates
- **Reproducibility (20%)**: Clean dataset saved correctly, code modular
- **Reflection (20%)**: Notebook comparisons and notes on tradeoffs

Example Expectations

- Code functions with docstrings
- Notebook demonstrating transformations

- Clear README documentation
- Saved cleaned dataset